BEFORE YOU READ

How far the SVD reaches

The spectral decomposition needs a symmetric matrix; the SVD works for every matrix. Moving outward, the two decompositions part ways.

$$S_+ \subset S \subset M_n \subset M_{m\times n}$$

$M_{m\times n}$ **All matrices**

adds rectangular shapes · only the SVD applies

M_n **Square**

eigendecomposition may not exist

S **Symmetric**

adds negative eigenvalues · $\sigma_i = |\lambda_i|$, signs absorbed into U or V

S_+ **Symmetric positive semi-definite**

spectral decomposition and SVD coincide: $U = V = Q, \Sigma = D$

TAKEAWAY

The broadest possible class

Where eigendecomposition may fail or not apply, the SVD always does: it covers non-symmetric and rectangular matrices too.